

1.7 Solve Linear-Quadratic Systems

Lesson Outline:

Part 1: Do It Now - review of substitution

Part 2: Possible solutions for a lin-quad system

Part 3: Solve linear-quadratic systems

Part 4: Application

DO IT NOW!

Solve the following linear system using the method of substitution:

① $y = 3x + 7$

② $y = 2x - 5$

$$3x + 7 = 2x - 5$$

$$3x - 2x = -5 - 7$$

$$x = -12$$

sub x -value back in to ① or ② and solve for y .

$$y = 3x + 7$$

$$y = 3(-12) + 7$$

$$y = -29$$

so the POI is $(-12, -29)$

Recall: solving a linear system means to find the point of intersection (POI)

Method of Substitution: solving a linear system by substituting for one variable from one equation into the other equation.

Steps to Solving A Linear-Quadratic System

1. Set equations equal to each-other

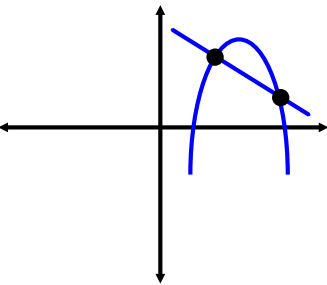
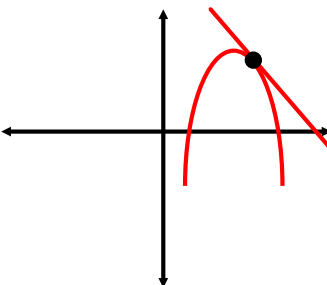
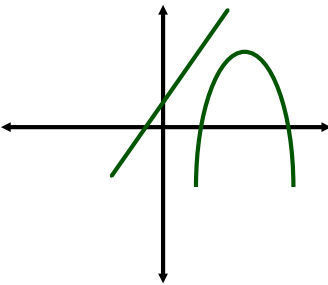
Line = Parabola

2. Rearrange to set the equation equal to zero

3. Solve for x by factoring or using the QF (the solution will tell you for what value of x the functions have the same y value)

4. Plug this value of x back in to either of the original functions to solve for y .

Possible solutions for a linear-quadratic system:

2 intersections	1 intersection	0 intersections
		
Secant: A line that intersects a curve at two distinct points. discriminant > 0	Tangent line: A line that touches a curve at one point and has the slope of the curve at that point. discriminant $= 0$	

Example 1

a) How many points of intersection are there for the following system of equations?

$$f(x) = \frac{1}{2}x^2 + 2x - 8 \quad g(x) = 4x - 10$$

$$\text{Set } f(x) = g(x)$$

$$\frac{1}{2}x^2 + 2x - 8 = 4x - 10 \quad (\text{set equal to each other})$$

$$\frac{1}{2}x^2 + 2x - 4x - 8 + 10 = 0 \quad (\text{set equal to zero})$$

$$\frac{1}{2}x^2 - 2x + 2 = 0$$

(common factor)

$$\frac{1}{2}(x^2 - 4x + 4) = 0$$

$$x^2 - 4x + 4 = 0$$

$$b^2 - 4ac = (-4)^2 - 4(1)(4)$$

$$= 0$$

(check discriminant)

∴ 1 solution

b) Solve the linear-quadratic system (give exact answers)

$$x^2 - 4x + 4 = 0$$

solve by factoring.

(hint: it is a perfect square trinomial)

$$(x-2)^2 = 0$$

$$x = 2$$

Plug $x=2$ back in to either original equation (linear is usually easier)

$$g(x) = 4x - 10$$

$$g(2) = 4(2) - 10$$

$$g(2) = -2$$

∴ The POI is $(2, -2)$

Example 2

Solve the following linear quadratic system

$$y = 3x^2 + 21x - 5$$

$$y = 10x - 1$$

$$3x^2 + 21x - 5 = 10x - 1$$

$$3x^2 + 11x - 4 = 0$$

$$(x+4)(3x-1) = 0$$

$$x+4=0 \quad 3x-1=0$$

$$x_1 = -4 \quad x_2 = \frac{1}{3}$$

POI #1

$$y = 10(-4) - 1$$

$$y = -41$$

$$(-4, -41)$$

POI #2

$$y = 10\left(\frac{1}{3}\right) - 1$$

$$y = \frac{10}{3} - \frac{3}{3}$$

$$y = \frac{7}{3}$$

$$\left(\frac{1}{3}, \frac{7}{3}\right)$$

Part 4: Application

Example 3: If a line with slope 4 has one point of intersection with the quadratic function $y = \frac{1}{2}x^2 + 2x - 8$, what is the y-intercept of the line? Write the equation of the line in slope y-intercept form.

$$4x + k = \frac{1}{2}x^2 + 2x - 8$$

$$0 = \frac{1}{2}x^2 - 2x - 8 - k$$

$$\text{Then } a = \frac{1}{2} \quad b = -2 \quad \text{and } c = -8 - k$$

$$b^2 - 4ac = 0$$

$$(-2)^2 - 4\left(\frac{1}{2}\right)(-8 - k) = 0$$

$$4 - 2(-8 - k) = 0$$

$$4 + 16 + 2k = 0$$

$$2k = -20$$

$$k = -10$$

∴ The equation of the line must be $y = 4x - 10$

Recall: equation of a line is $y = mx + k$ where k is the y-intercept and m is the slope.

Recall: for a lin-quad system to have 1 solution, the discriminant must be zero.